

Product Reference Manual (V1.0)

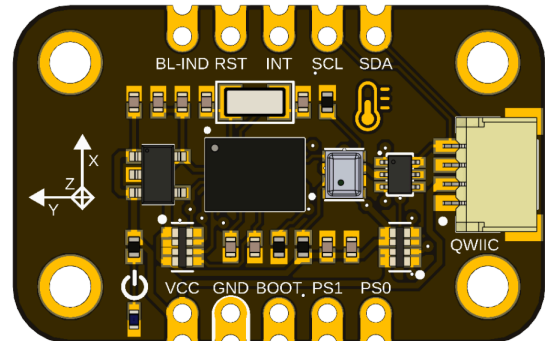
Description

The UNIT BNO055 + BMP280 module is a compact, fully integrated sensing solution designed to provide both motion tracking and environmental monitoring capabilities on a single board.

This module combines two high-performance sensors: the BNO055, a 9-axis absolute orientation sensor, and the BMP280, a high-precision barometric pressure and temperature sensor. The BNO055 integrates an accelerometer, gyroscope, and magnetometer, along with an embedded microcontroller that performs real-time sensor fusion. This enables the direct output of orientation data—including quaternions, Euler angles, and gravity vectors—eliminating the need for complex processing on the host system.

Complementing this functionality, the BMP280 delivers accurate atmospheric pressure and temperature measurements, making it suitable for applications such as altitude estimation and environmental monitoring.

By combining motion sensing and environmental data acquisition, the module provides a comprehensive solution for applications requiring spatial awareness and environmental context. It is particularly well-suited for use in robotics, unmanned aerial systems (drones), virtual and augmented reality platforms, and IoT-based sensing systems.



Key Features

- Integrated **9-axis absolute orientation sensor (BNO055)**
- Built-in **sensor fusion engine** with direct high-level data output
- Integrated **barometric pressure and temperature sensor (BMP280)**
- Simultaneous motion and environmental sensing
- High accuracy and stability for dynamic applications
- Reduced host MCU processing requirements
- Standardized **I²C communication interface**
- Fully compatible with **DevLab ecosystem**
- Plug-and-play integration via **Qwiic / STEMMA QT connectors**
- Compact form factor optimized for embedded applications

Hardware Features

- Dual-sensor architecture: BNO055 + BMP280
- Integrated onboard voltage regulation (depending on module design)
- I²C communication interface with configurable addressing
- Standard JST SH 1.0 mm 4-pin connector
- Support for daisy-chain I²C connections
- Compact PCB layout optimized for signal integrity
- Mounting-friendly design for prototyping and integration
- Low power consumption suitable for portable devices

Software Support

The module is designed for seamless integration with common embedded development environments:

- **Arduino IDE**
 - Compatible with widely used libraries for BNO055 and BMP280
- **ESP-IDF**
 - Suitable for ESP32-based systems requiring advanced control
- **MicroPython**
 - Rapid prototyping and scripting applications
- Example code and drivers available through:
 - UNIT Electronics repositories
 - DevLab ecosystem documentation

Typical software capabilities include the following:

- Orientation data acquisition (Euler angles, quaternions)
- Linear acceleration and gravity vector reading
- Atmospheric pressure and temperature measurement
- Altitude estimation algorithms
- Sensor calibration handling (BNO055)

Applications

- Robotics and autonomous systems
- Unmanned aerial vehicles (UAVs / drones)
- Virtual Reality (VR) and Augmented Reality (AR) systems
- Wearable motion tracking devices
- Environmental monitoring stations
- Industrial IoT (IIoT) sensing nodes
- Navigation and positioning systems
- Smart devices requiring orientation awareness

DevLab format compatibility.

Simplicity and compatibility are the primary objectives of the DevLab form factor. It offers a board layout that is compact and optimized for serial communication, which includes I²C and SPI interfaces.

This format enables the establishment of rapid and dependable connections via a Protocol I²C connector that is entirely compliant with the Qwiic and STEMMA standards.

This design guarantees efficient prototyping, accessibility, and simplicity of integration across a variety of devices and modules.

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1 The Board

1.1 Accessories

- 2x tiras header macho 2.54 mm (1x5 pines)
- 1x QWIIC 4-pin JST 1.0 mm Harness

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VCC	Supply Voltage	3.3	5.0	6.0	V
VIH	I ² C Input High Voltage	$0.7 \times 3.3V$	—	3.3v	V
VIL	I ² C Input Low Voltage	0	—	$0.3 \times VCC$	V
fSCL	I ² C Clock Frequency (Standard/Fast Mode)	—	100	400	kHz
TA	Operating Ambient Temperature	-40	25	+85	°C
RH	Relative Humidity (Non-condensing)	5	—	95	%
tPU	Power-Up Time Before Communication	—	650	—	ms
IAVG	Average Supply Current (Typical, Fusion Enabled)	—	12	18	mA

3. Functional Overview

The UNIT BNO055 + BMP280 module combines motion sensing and environmental monitoring capabilities on a single compact platform. The design integrates two specialized sensors connected through a shared I²C communication bus, allowing simultaneous acquisition of orientation, motion, temperature, and atmospheric pressure data.

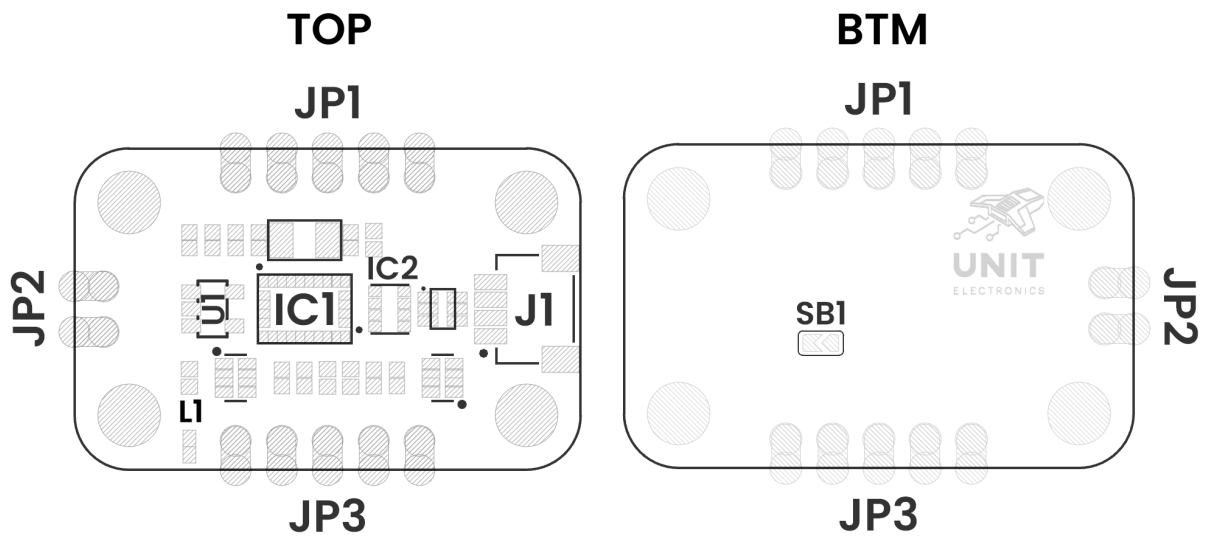
The BNO055 serves as the primary motion processing device. It integrates a 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer, together with an embedded microcontroller that performs real-time sensor fusion. This architecture allows the module to provide high-level outputs such as Euler angles, quaternions, gravity vectors, linear acceleration, and angular velocity without requiring complex calculations on the host processor.

The BMP280 complements the motion-sensing capabilities by providing accurate measurements of ambient temperature and atmospheric pressure. These measurements can be used for environmental monitoring and altitude estimation applications.

Both sensors communicate through the onboard I²C bus, which is routed to the JST-SH QWIIC/STEMMA-compatible connector and the expansion headers. This approach simplifies integration with a wide range of microcontroller platforms while reducing wiring complexity.

The combination of orientation sensing and environmental monitoring makes the module suitable for robotics, navigation systems, drones, wearable devices, industrial monitoring equipment, and IoT applications requiring both spatial awareness and environmental context.

3.2 Board Topology



Views of Board Topology

Views of Topology

Table 3.2.1 - Components Overview

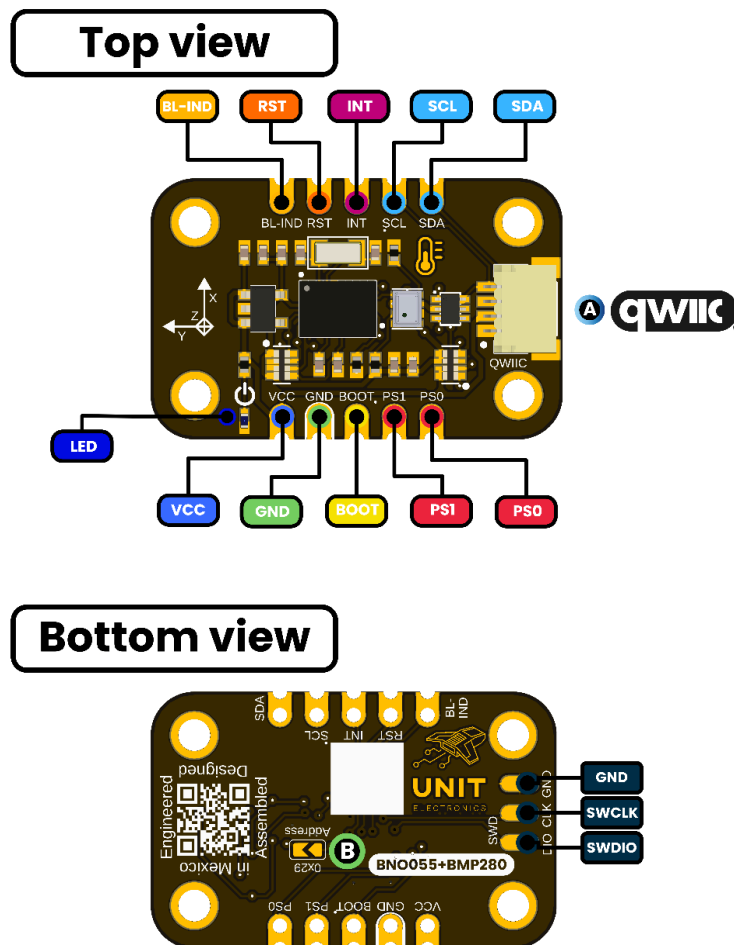
Ref.	Description
IC1	BNO055 9-axis absolute orientation sensor with integrated sensor fusion
IC2	BMP280 digital barometric pressure and temperature sensor
U1	AP2112K 3.3 V low-dropout voltage regulator
L1	Power-on indicator LED
SB1	BNO055 I ² C address selection solder bridge: open = 0x28, closed = 0x29
J1	Low-power I ² C QWIIC JST connector
JP1	2.54 mm BNO055 interface and control header: SDA, SCL, INT, nRESET and BL_IND
JP2	2.54 mm BNO055 SWD service/debug header: SWCLK and SWDIO
JP3	2.54 mm configuration header: VCC, GND, nBOOT, PS1 and PS0

4 Connectors & Pinouts

4.1 General Pinout

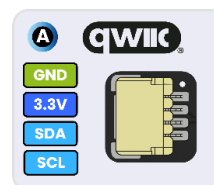
PINOUT

DevLab: I2C BNO055 + BMP280 Fusion Sensor



Description:

- | | |
|---|--|
| Supply voltage | Bootloader indicator |
| GND | I2C |
| Boot Mode | Interrupt |
| Protocol select | Reset |
| SWD | B Configurable address |



DevLab I2C BNO055 + BMP280 Fusion Sensor General Pinout

4.2 Pinout General Description

Pin	Function	Description
VCC	Power	Main module supply input (Max 6.0 V). The onboard AP2112K regulator generates the internal 3.3 V rail.
GND	Ground	Common ground reference for power and digital signals.
SDA	I ² C Data	Bidirectional I ² C data line shared by the BNO055, BMP280 and QWIIC connector.
SCL	I ² C Clock	I ² C clock line shared by the BNO055, BMP280 and QWIIC connector.
INT	Interrupt	Configurable interrupt output generated by the BNO055.
RST	Reset	Active-low hardware reset input for the BNO055. Pull low momentarily to reset the sensor.
BL-IND	Bootloader indicator	BNO055 bootloader-status output. Intended primarily for service or firmware-update operations.
BOOT	BOOT0	Active-low bootloader selection input for the BNO055. Hold low during reset or power-up to enter its internal bootloader.
PS0	Protocol Select 0	BNO055 communication-interface selection input.
PS1	Protocol Select 1	BNO055 communication-interface selection input.
SWCLK	SWD Clock	Debug clock for the internal BNO055 microcontroller. Intended for factory, service or advanced debugging use.
SWDIO	SWD Data	Bidirectional debug data for the internal BNO055 microcontroller. Intended for factory, service or advanced debugging use.
QWIIC	JST-SH I ² C Connector	Four-pin connector carrying GND, VCC, SDA and SCL. It is electrically connected to the same I ² C bus as the pin headers.

Note: Power the board from a **5 V VCC supply (recommended)**. Do not exceed the maximum rated input voltage. Configure the **PS0** and **PS1** pins to select the desired communication interface **before applying power or releasing the nRESET pin**.

4.3 Interface Configuration

Table 4.3.1 – Communication Interface Selection

PS1	PS0	Interface
0	0	Standard/Fast I ² C (Default)
0	1	HID-I ² C
1	0	UART
1	1	Reserved (Do not use)

Table 4.3.2 – I²C Address Configuration

Device	Address Selection	I ² C Address
BNO055	SB1 Open	0x28
BNO055	SB1 Closed (	0x29
BMP280	Fixed (SDO = GND)	0x76

Note: SB1 only changes the I²C address of the BNO055. The BMP280 is permanently configured at address 0x76.

Table 4.3.3 – Boot and Debug Signals

Signal	Direction	Description
BOOT	Input	Hold low during reset to enter the BNO055 bootloader.
RESET	Input	Active-low hardware reset.
BL_IND	Output	Bootloader status indication.

SWCLK	I/O	Serial Wire Debug clock (service/debug).
SWDIO	I/O	Serial Wire Debug data (service/debug).

Operational Notes

- A 5 V supply connected to VCC is recommended.
- Do not exceed the maximum rated input voltage.
- Configure PS0 and PS1 before applying power or releasing nRESET.
- The QWIIC connector and the SDA/SCL header pins are connected to the same I²C bus.
- The BNO055 and BMP280 operate internally at 3.3 V.
- SB1 only affects the BNO055 I²C address.

5 Board Operation

This example demonstrates the basic operation of the UNIT BNO055 + BMP280 module using the Arduino IDE. The firmware initializes both onboard sensors through the I²C interface and periodically acquires motion, orientation, temperature, and atmospheric pressure measurements, the BNO055 provides Euler orientation angles, gyroscope data, acceleration, linear acceleration, magnetic field measurements, gravity vectors, internal temperature, and calibration status. The BMP280 provides ambient temperature and barometric pressure measurements.

All acquired data is transmitted through the serial port in JSON format, allowing easy integration with data loggers, web applications, dashboards, IoT systems, or custom monitoring software.

Hardware Connection

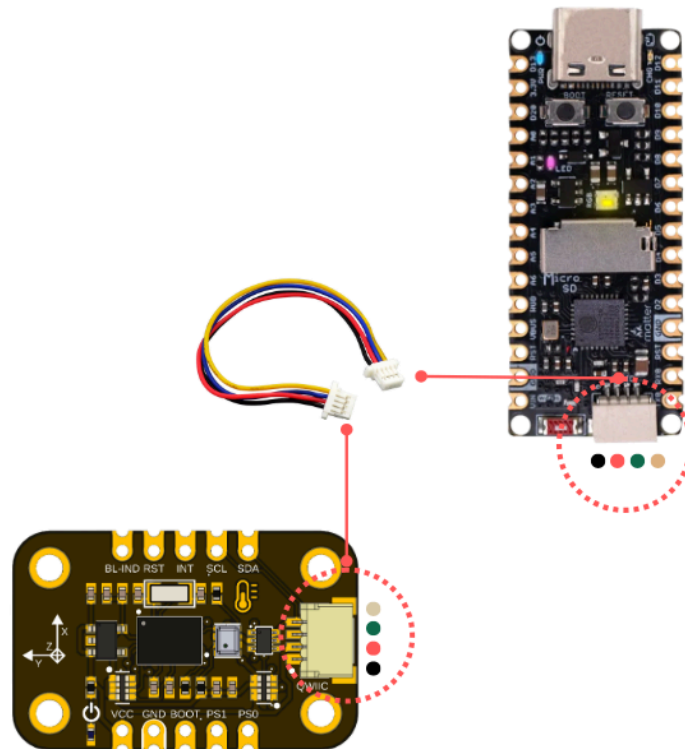
The module communicates through the I²C interface using the integrated JST-SH 1.0 mm DevLab/QWIIC/STEMMA-compatible connector. For the recommended setup, connect the module directly to the host controller using a standard 4-wire JST-SH cable.

In the example shown, the module is connected to an ESP32-C6 development board using:

Signal	PULSAR Pin
SDA	GPIO 6
SCL	GPIO 7
VCC	3.3 V
GND	GND

The JST-SH connector is the preferred connection method because it simplifies wiring, reduces connection errors, and provides compatibility with the QWIIC and STEMMA ecosystems. The same I²C signals are also available through the board headers for custom integrations.

After power is applied and the I²C bus is initialized by the host controller, both onboard sensors become accessible and can be queried for orientation, motion, temperature, and pressure measurements.



Example Code

```
#include <Wire.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_BNO055.h>
#include <Adafruit_BMP280.h>
#include <utility/imuMaths.h>

// BNO055: orientación y movimiento
Adafruit_BNO055 bno(55, 0x28, &Wire);

// BMP280: temperatura y presión
Adafruit_BMP280 bmp;
Adafruit_Sensor* bmp_temp = bmp.getTemperatureSensor();
Adafruit_Sensor* bmp_pressure = bmp.getPressureSensor();

void setup() {
  Serial.begin(115200);
  while (!Serial);

  // Inicializar BNO055
```

```

if (!bno.begin()) {
    Serial.println("{\\\"error\\\":\\\"BNO055 not detected\\\"}");
    while (1);
}
delay(1000); // Estabilizar BNO055

// Inicializar BMP280 en dirección 0x76
if (!bmp.begin(0x76)) {
    Serial.println("{\\\"error\\\":\\\"BMP280 not detected at 0x76\\\"}");
    while (1);
}

bmp.setSampling(Adafruit_BMP280::MODE_NORMAL,
                Adafruit_BMP280::SAMPLING_X2,
                Adafruit_BMP280::SAMPLING_X16,
                Adafruit_BMP280::FILTER_X16,
                Adafruit_BMP280::STANDBY_MS_500);
}

void loop() {
    // Eventos del BNO055
    sensors_event_t orientation, gyro, linearAccel, magnetometer, accel,
    gravity;

    bno.getEvent(&orientation,    Adafruit_BNO055::VECTOR_EULER);
    bno.getEvent(&gyro,           Adafruit_BNO055::VECTOR_GYROSCOPE);
    bno.getEvent(&linearAccel,    Adafruit_BNO055::VECTOR_LINEARACCEL);
    bno.getEvent(&magnetometer,   Adafruit_BNO055::VECTOR_MAGNETOMETER);
    bno.getEvent(&accel,          Adafruit_BNO055::VECTOR_ACCELEROMETER);
    bno.getEvent(&gravity,        Adafruit_BNO055::VECTOR_GRAVITY);

    // BMP280: temperatura y presión
    sensors_event_t temp_event, pressure_event;
    bmp_temp->getEvent(&temp_event);
    bmp_pressure->getEvent(&pressure_event);

    // Calibración BNO055
    int8_t temp_bno = bno.getTemp();
    uint8_t sys, gyroCal, accelCal, magCal;

```

```
bno.getCalibration(&sys, &gyroCal, &accelCal, &magCal);

// JSON OUTPUT
Serial.print("{");

// BMP280
Serial.print("\"bmp280\":{\"temperature\":");
Serial.print(temp_event.temperature, 2);

Serial.print(",\"pressure\":"); Serial.print(pressure_event.pressure,
2); Serial.print("},");

// BNO055 - orientación
Serial.print("\"orientation\":{\"x\":");
Serial.print(orientation.orientation.x, 2);

Serial.print(",\"y\":"); Serial.print(orientation.orientation.y, 2);
Serial.print(",\"z\":"); Serial.print(orientation.orientation.z, 2);
Serial.print("},");

// BNO055 - giroscopio
Serial.print("\"gyroscope\":{\"x\":"); Serial.print(gyro.gyro.x, 2);
Serial.print(",\"y\":"); Serial.print(gyro.gyro.y, 2);
Serial.print(",\"z\":"); Serial.print(gyro.gyro.z, 2);
Serial.print("},");

// BNO055 - aceleración lineal
Serial.print("\"linear_accel\":{\"x\":");
Serial.print(linearAccel.acceleration.x, 2);

Serial.print(",\"y\":"); Serial.print(linearAccel.acceleration.y, 2);
Serial.print(",\"z\":"); Serial.print(linearAccel.acceleration.z, 2);
Serial.print("},");

// BNO055 - magnetómetro
Serial.print("\"magnetometer\":{\"x\":");
Serial.print(magnetometer.magnetic.x, 2);

Serial.print(",\"y\":"); Serial.print(magnetometer.magnetic.y, 2);
Serial.print(",\"z\":"); Serial.print(magnetometer.magnetic.z, 2);
Serial.print("},");

// BNO055 - aceleración total
Serial.print("\"acceleration\":{\"x\":");
Serial.print(accel.acceleration.x, 2);
```

```
Serial.print(", \"y\":"); Serial.print(accel.acceleration.y, 2);
Serial.print(", \"z\":"); Serial.print(accel.acceleration.z, 2);
Serial.print("},");

// BNO055 - gravedad
Serial.print("\"gravity\":{ \"x\":");
Serial.print(gravity.acceleration.x, 2);
Serial.print(", \"y\":"); Serial.print(gravity.acceleration.y, 2);
Serial.print(", \"z\":"); Serial.print(gravity.acceleration.z, 2);
Serial.print("},");

// Temperatura interna y calibración
Serial.print("\"bno055_temp\":"); Serial.print(temp_bno);
Serial.print(",");
Serial.print("\"calibration\":{ \"sys\":"); Serial.print(sys);
Serial.print(", \"gyro\":"); Serial.print(gyroCal);
Serial.print(", \"accel\":"); Serial.print(accelCal);
Serial.print(", \"mag\":"); Serial.print(magCal); Serial.print("}");

Serial.println("}");
delay(500);
}
```

Expected Output

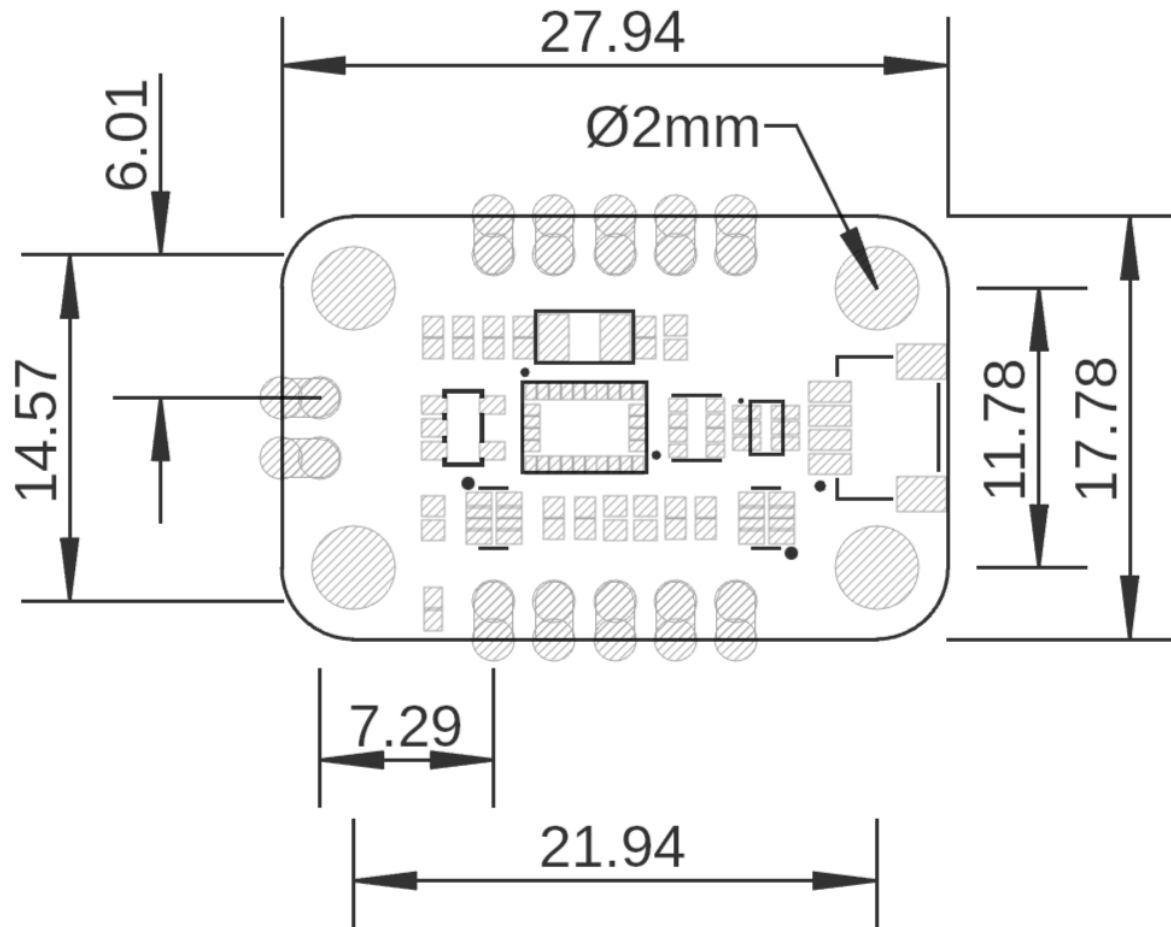
The serial monitor will display measurements in JSON format similar to the example shown below:

```
{
  "bmp280":{
    "temperature":25.40,
    "pressure":1012.35
  },
  "orientation":{
    "x":180.25,
    "y":2.10,
    "z":-1.50
  },
  "gyroscope":{
```

```
"x":0.02,  
"y":0.01,  
"z":0.00  
,  
"bno055_temp":28,  
"calibration":{  
  "sys":3,  
  "gyro":3,  
  "accel":3,  
  "mag":3  
}  
}
```

This example can be used as a starting point for robotics, navigation systems, environmental monitoring, motion tracking, and IoT applications requiring both orientation and environmental sensing.

6. Mechanical Information



Mechanical dimensions in millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	Link
Documentation	https://wiki.uelectronics.com/wiki/unit_devlab_i2c_bno055_bmp280_fusion_sensor
Github	https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_bno055_bmp280_fusion_sensor
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1 Schematic

(https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_bno055_bmp280_fusion_sensor/blob/main/hardware/unit_sch_v_1_1_0_ue0091_bno055_bmp280.pdf)

